RESEARCH PROPOSAL

Can an LLM understand a game tileset?

Benchmarking LLM Tile Understanding and Tile Use

Recognize tile relationships.

Reconstruct tilesets. Generate coherent levels.

Existing WFC/SAT research • A reproducible benchmark • Collaboration

Three questions for multimodal models

01 GENERATE

Given a tileset, can a model build a coherent level?

02 RELATE

Can it infer relationships among tiles and groups?

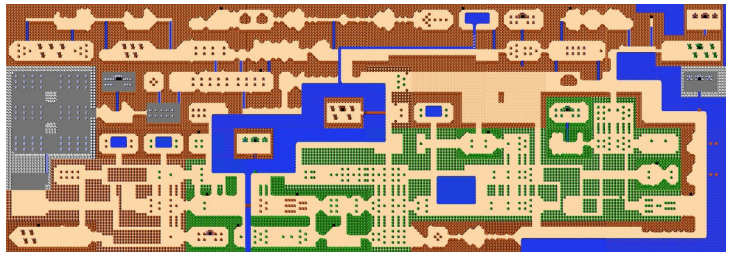
03 LEARN

How much do example levels improve performance?

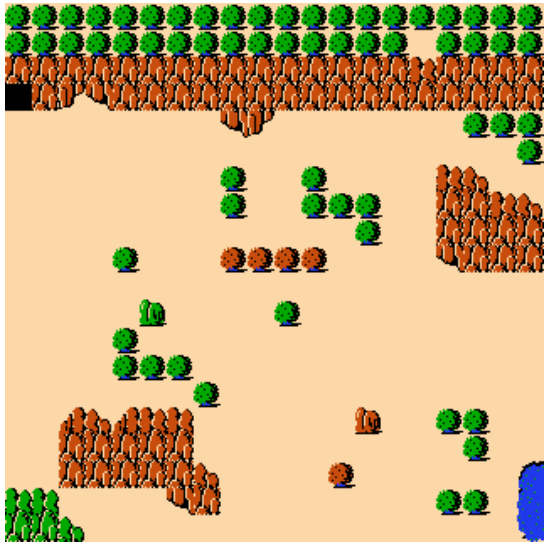
Zelda: original and generated levels

WFC and SAT motivate tests of higher-level structural understanding

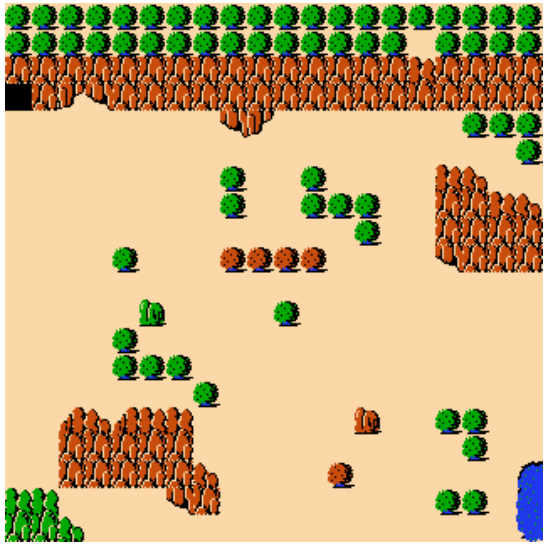
Original Zelda map



Context-sensitive WFC



Context-sensitive WFC-as-SAT



Existing algorithms can reproduce local patterns without recovering every semantic object.

Tiny Town: level image ⇄ tileset

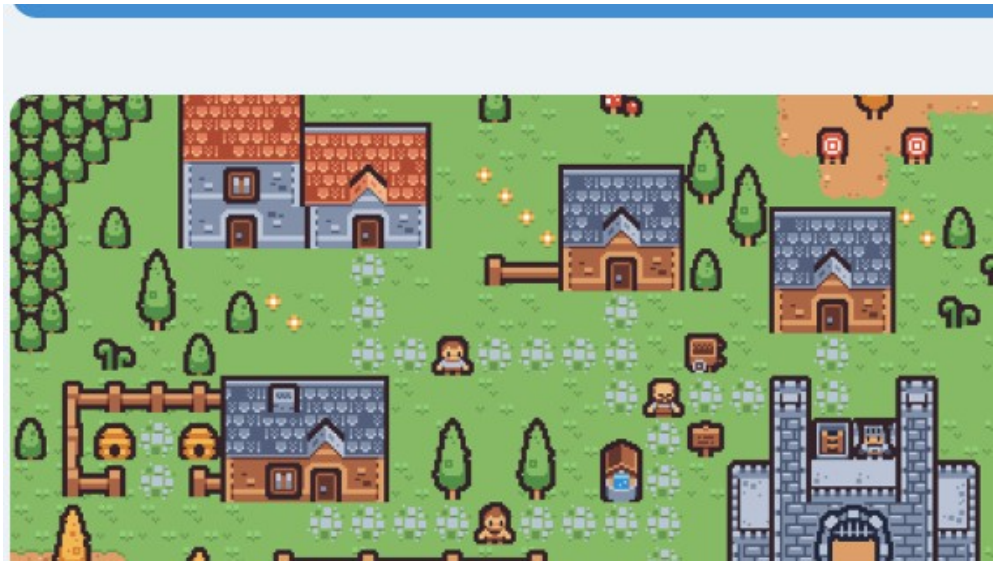
Two inverse tasks using the same source images • Kenney Tiny Town (CC0)

TILESET / PALETTE



Tileset → construct a coherent town

EXAMPLE TOWN IMAGE



Town image → recover its tileset

A tileset is more than a bag of images

What relationships should a multimodal model discover?

ROLE

Roof • wall • floor • door

LOCAL

Which tile edges can meet?

STRUCTURE

Which groups form a house or room?

Treat predictions as hypotheses; validate them against annotations and explicit rules.

Examples or no examples?

Hold the model, task, tile IDs, and scoring constant

A TILESET ONLY

Atlas + tile IDs + the same question or construction task.

B TILESET + LEVELS

Same atlas and task, plus one or more example levels.

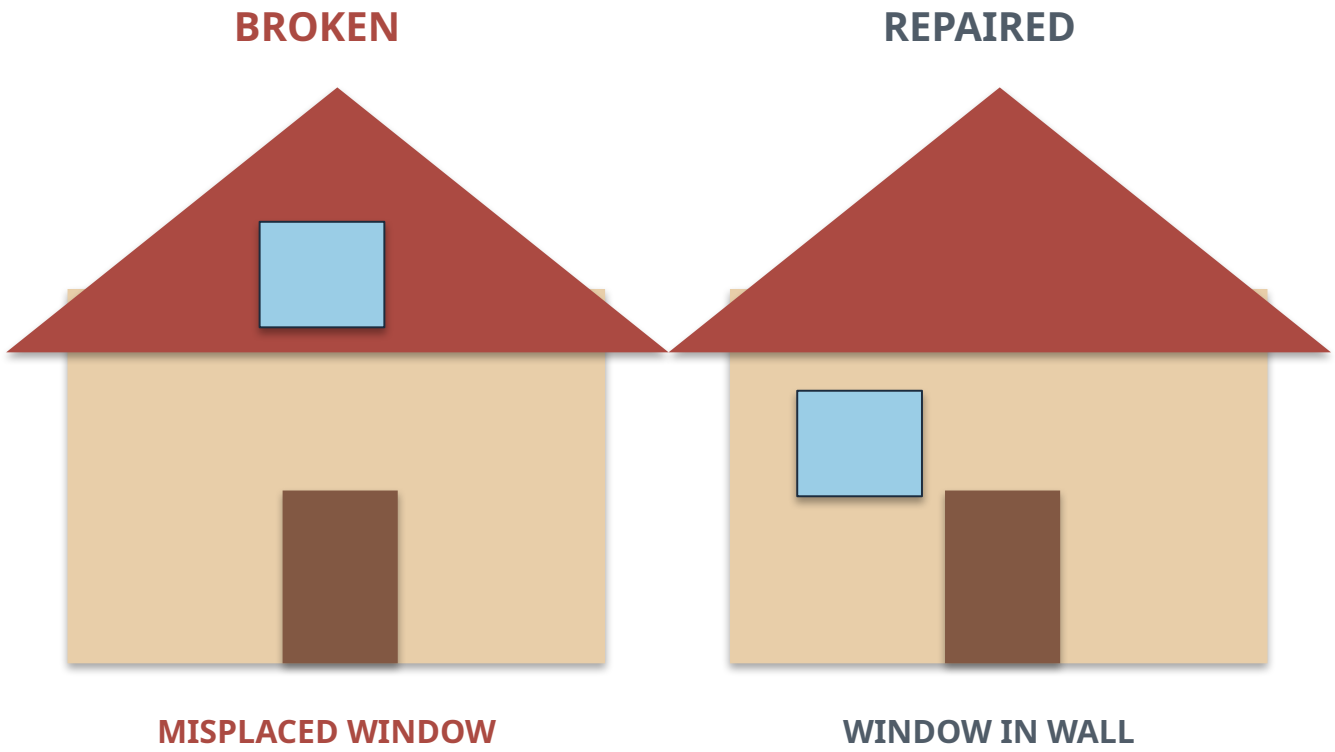
Candidates: Kenney Tiny Town, Tiny Dungeon, Tiny Farm, and Zelda.

Recognize, generate, and repair

Example: repair an invalid Tiny Town house arrangement

Four benchmark tasks

1. Identify tile roles
2. Predict adjacency and assemblies
3. Generate a tile-ID grid
4. Repair a broken arrangement



Repair test: give the model the damaged arrangement and the available tiles; ask it to identify and replace the invalid tile.

How do we measure success?

Objective scores for labeled tasks; blind human judgments for complete levels

Precision, recall, and F1

Precision: of predicted roof tiles, how many are correct?
Recall: of all true roof tiles, how many were found?

F1 = 2 × (precision × recall) / (precision + recall)

Example: precision 75%, recall 60% → F1 = 66.7%

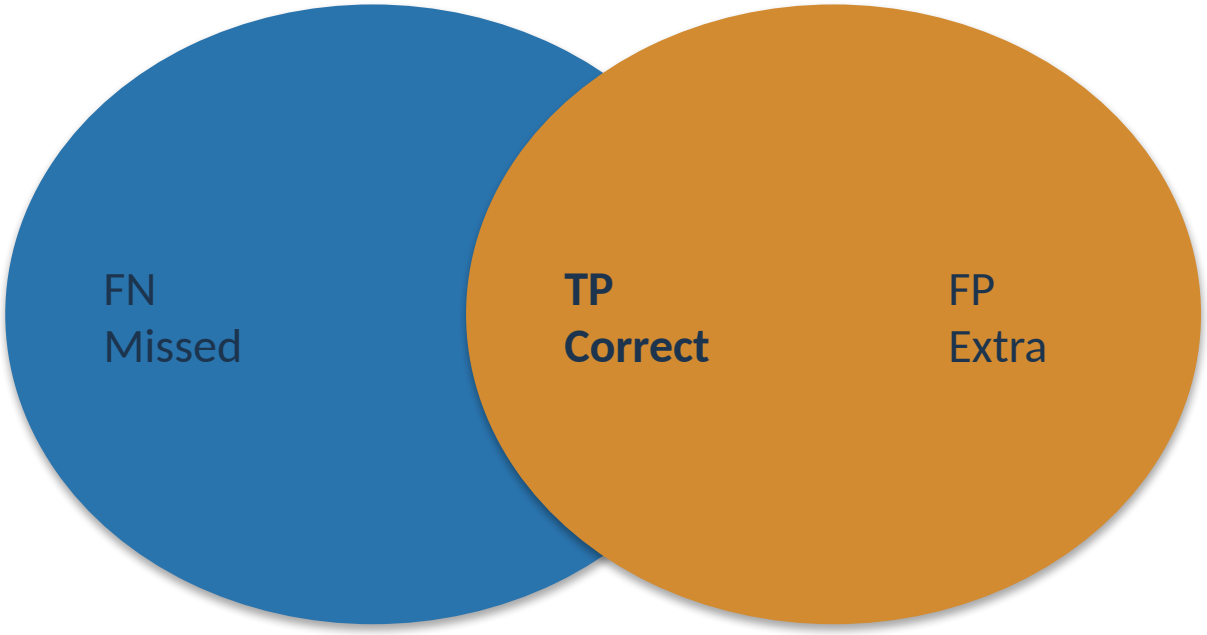
Blind human coherence rubric

Independent evaluators score generated levels without knowing which model or condition produced them.

Score each 1–5:
- Local tile connections
- Recognizable buildings and roads
- Overall level plausibility

Precision and recall as overlapping sets

Actual roof tiles (A) versus tiles predicted to be roofs (B)



Precision = $|A \cap B| / |B|$

Recall = $|A \cap B| / |A|$

Both measure the overlap, but use different denominators.

F1: harmonic mean and set overlap

A single score balances precision and recall

$$F1 = 2PR / (P + R) = 2 / (1/P + 1/R)$$

$$F1 = 2|A \cap B| / (|A| + |B|)$$

Example: $|A| = 8$, $|B| = 12$, $|A \cap B| = 6$
Precision = 6/12 = 0.50 Recall = 6/8 = 0.75

F1 = 12/20 = 0.60 (the Sørensen–Dice coefficient)

Fβ: choose how to weight precision and recall

β > 1 emphasizes recall; β < 1 emphasizes precision

$$F\beta = (1 + \beta^2)PR / (\beta^2P + R)$$

$$1/F\beta = [1/(1 + \beta^2)](1/P) + [\beta^2/(1 + \beta^2)](1/R)$$

β = ½: precision gets 4× the weight of recall
β = 1: equal weighting (F1)
β = 2: recall gets 4× the weight of precision

Set form: $F\beta = (1 + \beta^2)|A \cap B| / (|B| + \beta^2|A|)$

Probability interpretation

Condition on predicted positives or actual positives

Precision = Pr(A | B)
Recall = Pr(B | A)

$$F\beta = (1 + \beta^2)Pr(A \cap B) / [Pr(B) + \beta^2 Pr(A)]$$

Interpretation: the same overlap is measured against two reference populations.
Use Fβ when missing true tiles and adding false tiles have different importance.

COLLABORATION

A project others can join

First milestone: one tileset, annotated tasks, and two input conditions

CONTRIBUTIONS

Choose and annotate examples

Test prompts and models

Develop simple baselines

Review relevant papers

LATER POSSIBILITIES

Identify objects, shapes, textures

Confirm inferred structures

Pass structural rules to WFC or SAT

Four possible student papers

Research directions students could pursue independently

VISION AND GENERATION

1. Recover a tileset from a level screenshot

2. Measure how example levels improve generation

STRUCTURE AND VALIDATION

3. Infer higher-level structures: doors, roofs, houses

4. Validate generated levels with symbolic rules

Resources and next steps

A research proposal, not a completed literature review

Tilesets: Kenney Tiny Town • Tiny Dungeon • Tiny Farm

Zelda map data: inventwithpython.com

Related work: Video Game Level Corpus • Tile Embedding Patterns and PCG • GameTileNet • Learned Level Representations

github.com/stevex24/tile-understanding-benchmark

Next: select a tileset, annotate ground truth, and run baseline trials.

Three papers to start with

Datasets, learned tile semantics, and constraint-based generation

01 THE VIDEO GAME LEVEL CORPUS

Summerville, Snodgrass, Mateas & Ontañón (2016)

A reusable dataset for evaluating game-level analysis and generation.

https://arxiv.org/abs/1606.07487

02 TILE EMBEDDING: A GENERAL REPRESENTATION FOR LEVEL GENERATION

Jadhav & Guzdial (AIIDE 2021)

Learning visual and semantic representations of tiles; directly relevant to tile understanding.

https://doi.org/10.1609/aiide.v17i1.18888

03 STURGEON: TILE-BASED PROCEDURAL LEVEL GENERATION

Cooper (AIIDE 2022)

Combines learned and designer-specified constraints for generation and repair.

https://doi.org/10.1609/aiide.v18i1.21944